

Ensemble methods

We consider data from a source space $\mathcal{X}$ and label space $\mathcal{Y}$, $l: \mathcal{Y}^2 \rightarrow \mathbb{R}^+$ a loss function, a sample is denoted $(x, y) \in \mathcal{X} \times \mathcal{Y}$, and its distribution is denoted $p(x, y)$ or p for short. Let us consider a neural network $f(\cdot, \theta): \mathcal{X} \rightarrow \mathcal{Y}$, with parameters θ . The objective of training consists in minimizing the generalization error defined as $\mathcal{L}(f, \theta) = E_{(x, y) \sim p}[l(f(x, \theta), y)]$. For that the network will be trained on a dataset D of size N , sampled from p using a training configuration c that may include different sources of randomness such as hyperparameters. We denote $e = (D, c)$ the corresponding learning procedure, $\theta(e) = \arg \min_{\theta} \frac{1}{N} \sum_{(x, y) \in D} l(f(x, \theta), y)$ denotes the weights learned through e . The generalization error of functions learned under the learning procedure $e \sim p_e$, with p_e a distribution on e , is defined as $\mathcal{L}_e(f) = E_{e \sim p_e}[\mathcal{L}(f, \theta(e))]$. Being an expectation, $\mathcal{L}_e(f)$ does not depend on a particular training set D or configuration c . It characterizes the performance of the class of estimators learned through e . In the following we will consider real valued functions f and MSE loss i.e. $l(f(x, \theta), y) = (f(x, \theta) - y)^2$.

1. Show that

$$\mathcal{L}_e(f) = E_{e \sim p_e}[\mathcal{L}(f, \theta(e))] = E_{(x, y) \sim p}[\text{bias}^2(f|(x, y)) + \text{var}(f|x)] \quad (1)$$

With $\text{bias}(f|(x, y)) = y - \bar{f}(x)$, $\text{var}(f|x) = E_e[(f(x, \theta(e)) - \bar{f}(x))^2]$, $\bar{f}(x) = E_e[f(x, \theta(e))]$

We now consider an ensemble estimator defined as $f_{\text{ens}}(\cdot, \theta_{1:M}) = \frac{1}{M} \sum_{m=1}^M f(\cdot, \theta_m)$ where $\theta_{1:M} = \{\theta_1, \dots, \theta_M\}$ and each θ_m or equivalently $f(\cdot, \theta_m)$ is a sample from the learning procedure. Let us denote $e_{1:M} = \{e_1, \dots, e_M\}$, so that the expectation over the distribution of the training procedures $e_1 \dots e_M$ writes as $E_{e_{1:M}}[\cdot] = E_{e_M} \left[E_{e_{M-1}} \left[\dots E_{e_1}[\cdot] \right] \right]$.

Let us define the bias and variance for f_{ens} :

$$\text{bias}(f_{\text{ens}}|(x, y)) = y - E_{e_{1:M}} \left[\frac{1}{M} \sum_{m=1}^M f(x, \theta_m) \right]$$

$$\text{var}(f_{\text{ens}}|x) = E_{e_{1:M}} \left[\left(\frac{1}{M} \sum_{m=1}^M f(x, \theta_m) - E_{e_{1:M}} \left[\frac{1}{M} \sum_{m=1}^M f(x, \theta_m) \right] \right)^2 \right]$$

1.1 Preliminaries, show the following two identities:

- $E_{e_{1:M}} \left[\frac{1}{M} \sum_{m=1}^M f(x, \theta_m) \right] = \frac{1}{M} \sum_{m=1}^M E_{e_m} [f(x, \theta_m)]$
- $(\sum_{m=1}^M a_m - b_m)^2 = \sum_{m=1}^M (a_m - b_m)^2 + \sum_{m=1}^M \sum_{m' \neq m} (a_m - b_m)(a_{m'} - b_{m'})$

Then using (1) for f_{ens} , show:

$$\mathcal{L}_{e_{1:M}}(f_{\text{ens}}) \triangleq E_{e_{1:M}}[\mathcal{L}(f_{\text{ens}}, \theta_{1:M})] = E_{(x, y) \sim p} \left[B^2 + \frac{1}{M} V + \frac{M-1}{M} C \right]$$

Where (for simplification in the following $f_m(\cdot)$ denotes $f(\cdot, \theta_m)$):

$$B = \frac{1}{M} \sum_{m=1}^M \text{bias}(f_m|(x, y)), \quad V = \frac{1}{M} \sum_{m=1}^M \text{var}(f_m|x),$$

$$C = \frac{1}{M(M-1)} \sum_m \sum_{m' \neq m} \text{cov}(f_m, f_{m'}|x) \quad \text{with} \quad \text{cov}(f, f'|x) = E_{e, e'}[(f - E_e[f])(f' - E_{e'}[f'])]$$

1.2 Let us now suppose that the f_m are identically distributed. This means that the bias, variance and expectation are the same for all the f_m . Let us denote respectively $\text{bias}(f|(x, y))$, $\text{var}(f|x)$, $\bar{f}$ these different variables – defined as above (question 1). Show that:

$$E_{e_{1:M}}[\mathcal{L}(f_{\text{ens}}, \theta_{1:M})] = E_{(x, y) \sim p} \left[\text{bias}^2(f|(x, y)) + \frac{1}{M} \text{var}(f|x) + \frac{M-1}{M} \text{cov}(f, f'|x) \right] \quad (2)$$

Where $\text{cov}(f, f'|x) = E_{e, e'}[(f(x, \theta(e)) - \bar{f}(x))(f(x, \theta(e')) - \bar{f}(x))]$.

1.3 Interpret this last result. What should be the property of the f functions for minimizing this generalization error?

1.4 We will now suppose that the f_m are independent, meaning that $\text{cov}(f, f'|x) = 0$.

What is the expression of $E_{e_{1:M}}[\mathcal{L}(f_{ens}, \theta_{1:M})]$?

What is the benefit of using an ensemble method compared to a single function estimator f ?

2. We will now consider an alternative to building ensembles, that emerged with the use of large pre-trained networks. We start from a pre-trained network (for example on ImageNet). Then we fine tune it on a given dataset using some learning procedure e . For different but related learning procedures e , one will get different set of parameters $\theta(e)$ that will be close one to the other. Suppose we fine tune M functions $f_m, m = 1 \dots, M$. Let us define the ensemble function:

$$f_{wa} \triangleq f(\cdot, \theta_{wa}) \text{ with } \theta_{wa} = \frac{1}{M} \sum_{m=1}^M \theta_m$$

This means that the ensemble is defined by a unique function f_{wa} , its weights being the average of the weights of the M functions f_m . θ_{wa} is supposed close to $\theta_m, \forall m$. The f_m are supposed identically distributed so that (2) applies.

2.1 Using a first order Taylor expansion of $f(\cdot, \theta_m)$ around $f(\cdot, \theta_{wa})$, show that

- $f_{ens} - f_{wa} = O(\Delta_{1:M}^2)$ with $\Delta_{1:M} = \max_m \|\theta_m - \theta_{wa}\|_2$

2.2 Using a zeroth order Taylor expansion w.r.t. its first argument of $l(f_{ens}(x), y)$ around $f_{wa}(x)$, show that

- $l(f_{ens}(x), y) = l(f_{wa}(x), y) + O(\Delta_{1:M}^2)$

2.3 Using this result show that

- $\mathcal{L}(f_{wa}, \theta_{1:M}) = \mathcal{L}(f_{ens}, \theta_{1:M}) + O(\Delta_{1:M}^2)$

Show then

- $E_{e_{1:M}}[\mathcal{L}(f_{wa}, \theta_{1:M})] = E_{e_{1:M}}[\mathcal{L}(f_{ens}, \theta_{1:M})] + O(\bar{\Delta}^2)$ with $\bar{\Delta}^2 = E_{e_{1:M}}[\Delta_{1:M}^2]$

2.4 Interpret this result. What could be the benefit of the f_{wa} estimator compared to the f_{ens} estimator?

Note - Taylor expansion. Let $f: \mathbb{R}^p \rightarrow \mathbb{R}$ a differentiable function, the Taylor expansion of order 1 around point a is: $f(a+h) = f(a) + \nabla f(a) \cdot h + O(\|h\|^2)$